MILITARY SPECIFICATION

POLES, STRUCTURAL WOODEN, FOR HIGH FREQUENCY (HF) ANTENNAS

1. SCOPE

1.1 Scope. This specification covers wooden poles used as structural supports or members for HF antennas, when used in conjunction with a detailed antenna specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-P-14527 - Pole Line Hardware, Packaging of

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

05.1-1963 Wood Poles, Specifications and Dimensions for

(Application for copies should be addressed to the ANSI 1430 Broadway, New York, New York 10018.)

AMERICAN WOOD PRESERVERS ASSOCIATION (AWPA)

C1 - All Timber Products Preservative Treatment by Pressure Processes
C4 - Poles Preservative Treatment by Pressure Processes
M1 - Standard for the Purchase of Treated Wood Products
M2 - Standard Instructions for the Inspection of Preservative Treatment of Wood
P1 - Standard for Creosote

(Application for copies should be addressed to the AWPA, 1012 14th St. N.W., Washington, D.C. 20005.)

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AMERICAN SOCIETY OF TESTING MATERIALS (ASTM)

A36 - Specification for Structural Steel
A153 - Specification for Zinc Coating (Hot Dip) on Iron and Steel Hardware.

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 General. Wooden poles shall conform to the applicable requirements of ANSI 05.1, and shall be Douglas Fir or Southern Pine.

3.2 Physical requirements.

3.2.1 Size. Unless otherwise specified in the procurement document (see 6.2), the size of the pole shall be as required to satisfy the structural requirements of its application, limitations due to the method or route of shipment and shall be a preferred standard size as specified in ANSI Specification 05.1 for the wood selected. Poles of a numerical class higher than 4 shall not be used (see 6.4.1).

3.2.1.1 Length. The length of the pole as measured between its extreme ends shall be at least the length specified in 3.2.1, (the branded nominal length) but may not exceed this length by more than 6 inches.

3.2.1.2 Diameter. The minimum and maximum diameters of the butt and top of the pole when calculated from measurements made by means of a circumference tape, shall neither be less than the diameters fixed by its length and class in ANSI 05.1 for the particular wood selected, (as branded on the pole) nor shall they exceed those diameters by more than 1 inch.

3.2.2 Limits of imperfection.

3.2.2.1 Seasoning checks. Seasoning checks of any length are permitted provided that they do not extend in depth beyond the pitch center of the pole. Splits or grain separation which extend from one surface to another are not permitted to extend more than 6 inches along the length of the pole and shall be limited to either butt or top ends.

3.2.2.2 Crooks. Poles shall be free from short crooks. Any sweep shall be in one plane and one direction only. A straight line drawn from a top edge to a bottom edge of a pole shall, at no point along the pole, fall more than 1 inch for every 10 feet of the pole length from the pole surface.

3.2.2.3 Spiral grain. The spiral grain (twist grain) shall not exceed the following limitations:

Length of pole	Maximum Twist of Grain
30 feet and shorter	1 complete twist in any 10 feet
30 feet to 45 feet incl.	1 complete twist in any 16 feet
50 feet and larger	1 complete twist in any 20 feet

3.2.3 Roof. A one-way roof, cut at an angle of 15 to 20 degrees from the horizontal, shall be provided the top of poles.

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3.2.4 Ground line. A ground line that will remain clearly visible up to the time of installation shall be marked on each pole. The ground line shall be so related to the radiating curtain and guy fastening positions as to assure maintenance of the electrical and mechanical characteristics of the antenna when the pole is buried to the ground line depth when installed on a level ground plane.

3.2.5 Conditioning prior to treatment. Prior conditioning shall be provided in accordance with the applicable requirements of AWPA standards C1, C4 and M1. Air seasoning is the preferred method of conditioning where climatic conditions and time are not unfavorable factors.

3.2.6 Preservative treatment. Preservative treatment by full cell pressure process shall be provided in accordance with the applicable requirements of AWPA standards C1 and C4. All possible pole cutting, boring, framing, sawing, trimming, marking, branding and making of inserts shall be done prior to pressure preservative treatment. Final steam bath is not permitted.

3.2.6.1 Preservative. The preservative shall be a creosote-coal tar. The creosote shall meet the requirements of the AWPA standard P1.

3.2.6.2 Final retention. Final retention shall not be less than 12.0 pounds per cubic foot for Douglas Fir and 9.0 pounds per cubic foot for Southern Pine.

3.2.7 Marking of poles. Poles shall bear a clear imprint of the two marks indicating acceptance by the inspector before and after treatment.

3.2.8 Instructions. Instructions shall be provided for field treatment of any accidental damage, field cutting, or boring which is unavoidable, by soaking the area with a minimum of three applications of hot preservative, or in the case of holes, by inserting a plug at one end and a pressure gun at the other, to force the preservative into the wood. The after-treatment preservative and pressure shall be the same as that originally used on the whole member. When specified in the procurement document, kits to accomplish the above repair shall also be supplied on a per destination basis (see 6.2). Preservative quantities shall be one quart per pole supplied.

3.3 Pole line hardware. All pole line hardware shall be of a material, strength, and grade as determined by the antenna design requirements.

3.3.1 Pole splicing. Where antenna pole heights exceed the economical and convenient lengths for transportation and handling purposes, antenna poles may be spliced. The method and location of splicing shall be governed by the antenna loadings and configuration. The splice shall be proportioned for the maximum bending, shear and direct stresses produced at the splice point. The pole splice shall provide the same strength as for an unspliced single pole. The design of the splice shall provide a method of field adjustment to insure a rigid and tight fit around the poles, preventing the possibility of movement of the pole within the splice. All hardware employed for the splice shall be zinc coated hot dipped galvanized in accordance with ASTM A153.

3.3.2 Steps. Standard hook pole steps, 5/8 inches diameter by 9 or 10 inches long, zinc-coated hot dip galvanized in accordance with ASTM A153, shall be provided for installation at the field site. Sufficient pole steps shall be provided to allow for an 18 inch spacing between steps (36 inches between adjacent steps on each side of pole), with the lowest step no lower than 6 feet 6 inches from the finished ground surface. The pole shall be predrilled with undersized holes for ease of pole step installation in the field.

3.3.3 Attachment hardware. Any attachment hardware necessary to connect the antenna to the pole shall be supplied, for field installation, and shall be zinc coated hot dipped galvanized in accordance with ASTM A153.

3.4 Anchorage. Wood pole base anchorage design shall avoid the use of through bolts. Where the use of through bolts is unavoidable, the bolts shall maintain sufficient edge distance from the pole butt to avoid pole splitting. In no case shall the bolt hole edge distance be less than 12 inches.

3.5 Wooden appendages. All wooden appendages such as crossarms, struts, and so forth shall be of the same wood and treatment as the associated pole.

3.6 Electrical requirements. The pole shall withstand the electric field environment produced by its associated antenna (see 6.4.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Supplier's quality assurance system. The supplier shall provide and maintain a documented quality assurance system acceptable to the Government for supplies and services covered by this specification (see 6.3).

4.1.1.1 Quality plan. The supplier shall prepare a quality plan encompassing all quality assurance actions to be utilized in controlling and verifying compliance to contract and specification requirements. The quality plan shall be submitted to the command or agency concerned for evaluation of acceptability prior to the suppliers quality conformance inspection (see 6.3).

4.1.2 Government verification. All quality assurance operations performed by the supplier will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written quality plan are being properly applied, b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the supplier to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test procedures. Test procedures shall be in accordance with AWPA Specifications except as otherwise specified herein.

4.1.4 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The method of examination and testing of the equipment fall within the following classifications:

(a) Quality conformance inspection (see 4.3)

(1) Production inspection (see 4.3.1)
(2) Production control inspection (see 4.3.2)
(3) Electrical verification (see 4.3.3)

(b) Inspection of preparation for delivery (see 4.4)

4.3 Quality conformance inspection.

4.3.1 Production inspection.

4.3.1.1 Inspection prior to treatment. All poles and appendages shall be given a detailed inspection for seasoning, size, location and dimensions of all cutting or machine operations, marking, branding, and so forth (see 3.2.1 through 3.2.4, 3.2.7, 3.3.2, 3.4, and 3.5).

4.3.1.2 Inspection after treatment. Inspection of poles after treatment shall be in accordance with AWPA standard M2. Preservative retention shall be determined by assay. When 15 percent or more of the poles inspected fail to meet the penetration requirements, all poles in the charge shall be rejected and retreated in accordance with AWPA standard C1. If less than 15 percent of the poles tested fail to pass, the nonconforming poles shall be retreated and all conforming poles accepted. Sample bore holes shall be treated and plugged in accordance with AWPA standard M2. Ground-Line marking shall meet the requirements of 3.2.4.

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4.3.1.3 Inspection of field damage treatment kits. Kits shall be inspected for adequate instructions and materials as necessary.

4.3.1.4 Pole line hardware inspection. Pole splices, steps, and attachment hardware shall be inspected to verify material (ASTM A36), and galvanizing (ASTM A153) requirements. A representative sample of splice hardware and poles shall be assembled and checked for fit.

4.3.2 Production control inspection.

4.3.2.1 Preservative inspection. Samples of the preservative shall be taken and tested in accordance with AWPA standard M2.

4.3.2.2 Process control inspection. Plant equipment and instrumentation shall be inspected in accordance with AWPA standard M2. Instrumentation shall show evidence of adequate periodic calibration.

4.3.3 Electrical verification. A representative sample of the treated poles shall be subjected to the worse case electric field, as determined by the supplier and agreed to by the command or agency concerned to determine compliance with 3.6.

4.3.4 Inspection reports. A report in accordance with AWPA standard M2 shall be available for command or agency concerned review at time of antenna acceptance inspection (see 6.3).

4.4 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by suppliers shall be specified in the individual order.)

5.1 Poles.

5.1.1 Preservation and packaging. Not applicable.

5.1.2 Packing. Packing or method of shipment of poles shall comply with Uniform Freight Classification rules or regulations of other carrier rules as applicable to the mode of transportation.

5.2 Pole line hardware. Pole line hardware (splices, steps and attachment hardware) shall be preserved and packaged level A or C and packed level A, B or C in accordance with MIL-P-14527 (see 6.2).

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The poles specified herein are intended for use as structural supports for HF antennas such as the Log Periodic, Inverted Cone, Rhombic and High Angle Types, under intense electromagnetic fields associated with these antennas. Installations will be world-wide, including the Tropics, Arctic, and the coastal water regions.

6.2 Ordering data. The specification is not complete unless used in conjunction with a detailed antenna specification. Procurement documents should specify the following:

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(a) Title, number and date of this specification.
(b) Title, number and date of the antenna specification, or the antenna type.
(c) Selection of applicable level of packaging and packing for pole line hardware (see 5.2).
(d) Size, if specifically desired (see 3.2.1).
(e) Limitations due to method or route of shipment, if any (see 3.2.1).
(f) Repair kit quantities and destinations, if desired (see 3.2.8).

6.3 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to the following:

(a) Supplier's quality assurance system (see 4.1.1).
(b) Quality plan (see 4.1.1.1).
(c) Inspection reports (see 4.3.4)

6.4 Definitions.

6.1 Size. Size is defined as the length and class of the pole, and any other dimensions determined by the length and class, such as to comply with ANSI 05.1 for the particular wood used (see 3.2.1).

6.4.2 Withstand. Withstand is defined as not being consumed, damaged, or destroyed due to dielectric heating or other electrical causes (see 3.6).

Preparing activity:
Navy - EC

(Project 5510-N135)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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NAVAL ELECTRONIC SYSTEMS COMMAND
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